

INR Datasets for Function-Space PCA

Survey of publicly available SIREN / neural field collections

Learning Projections Project

April 2026

Contents

1	Motivation	2
2	Datasets: Standalone INRs (independently queryable)	2
2.1	DWSNets MNIST / FashionMNIST	2
2.2	Implicit-Zoo CIFAR-10 / CityScapes / ImageNet	2
2.3	NFN CIFAR-10 INRs	3
2.4	Neural Field Arena (CVPR 2024)	3
2.5	COIN-Collection (MS-COCO, HuggingFace)	3
3	Datasets: Shared-backbone + modulation (not standalone)	4
3.1	Functa — CelebA-HQ-64 (DeepMind, ICML 2022)	4
3.2	MedNF / MedFuncta (550 k+ medical images, Zenodo 2025)	4
4	Summary Table	4
5	Recommendations for Function-Space PCA	4
6	Implementation in this Codebase	5

Motivation

Function-space PCA (FPCA) requires a dataset of functions that share a common domain. Implicit Neural Representations (INRs / SIRENs) are a natural source: each network encodes one image and can be evaluated at any continuous coordinate $(x, y) \in [0, 1]^2$. This report surveys all publicly available collections of pre-trained 2-D image SIRENs.

Two datasets are ready for immediate use with this codebase:

- **DWSNets MNIST/FashionMNIST** — 70 000 independent SIRENs, stacked `.pt` format, ~200 MB download.
- **Implicit-Zoo CIFAR-10** — 60 000 independent SIRENs, individual state-dict `.pt` files, ~1.4 GB download.

Datasets: Standalone INRs (independently queryable)

These datasets store one fully-independent MLP per image, queryable at any (x, y) .

DWSNets MNIST / FashionMNIST

Paper *Equivariant Architectures for Learning in Deep Weight Spaces*, Navon et al., ICML 2023.

Download https://www.dropbox.com/sh/56pakaxe58z29mq/AABtWNkRYroLYe_cE3c90DXVa?dl=0

Files: `mnist-inrs.zip`, `fmnist_inrs.zip`

GitHub <https://github.com/AvivNavon/DWSNets>

Images MNIST and FashionMNIST (28×28 grayscale)

Count 70 000 SIRENs per dataset (60 k train + 10 k test)

Architecture $2 \rightarrow 32 \rightarrow 32 \rightarrow 1$, SIREN, $\omega_0 = 30$

Format Single stacked `.pt` file: dict with keys `weights` (list of tensors (N, out, in)), `biases` (list of tensors (N, out)), `labels` ((N,)).

License MIT

Notes The community standard benchmark for weight-space learning. Also used by NFN, ScaleGMN, ARC, neural-graphs, and many others.

Quick-start (after download):

```
unzip mnist-inrs.zip -d data/dwsnets/  
python fpca.py +experiment=dwsnets_mnist_eulerian
```

Implicit-Zoo CIFAR-10 / CityScapes / ImageNet

Paper *Implicit-Zoo: A Large-Scale Dataset of Neural Implicit Functions for 2D Images and 3D Scenes*, Qi Ma et al., NeurIPS 2024.

Download (Kaggle) <https://www.kaggle.com/datasets/qimaqi/implicit-zoo-cifar10-inrs>
`kaggle datasets download -d qimaqi/implicit-zoo-cifar10-inrs`

GitHub <https://github.com/qimaqi/Implicit-Zoo>

Images CIFAR-10, CityScapes, ImageNet-100/1K

Count CIFAR-10: 60 000 SIRENs (1.4 GB); ImageNet-1K: 1.43 M (750 GB)

Architecture CIFAR-10: $2 \rightarrow 64 \rightarrow 64 \rightarrow 3$, SIREN, $\omega_0 = 30$

Format One `.pt` state dict per image, organized in class subdirectories. Supported key schemes: `layers.N.weight`, `net.N.linear.weight`.

License MIT

Notes The largest released collection of independent 2-D SIRENs. Start with CIFAR-10 (~1.4 GB); ImageNet-1K requires 750 GB.

Quick-start (after download):

```
unzip implicit-zoo-cifar10-inrs.zip -d data/implicit_zoo/  
python fpca.py +experiment=implicit_zoo_cifar_eulerian
```

NFN CIFAR-10 INRs

Paper *Permutation Equivariant Neural Functionals*, Zhou et al., NeurIPS 2023.

Download Google Drive link in repo README.

GitHub <https://github.com/AllanYangZhou/nfn>

Images CIFAR-10, 60 000 SIRENs

Architecture $2 \rightarrow 64 \rightarrow 64 \rightarrow 3$, SIREN, $\omega_0 = 30$ (same as Implicit-Zoo CIFAR-10)

Format Same stacked `.pt` format as DWSNets.

License MIT

Notes Cross-compatible with DWSNets loader; interchangeable with Implicit-Zoo CIFAR-10.

Neural Field Arena (CVPR 2024)

Paper *How to Train Neural Field Representations: A Comprehensive Study and Benchmark*, Papa et al., CVPR 2024.

Download <https://zenodo.org/records/10392793> (compressed ~1 GB; full ~127 GB)

GitHub <https://github.com/samuelepapa/neural-field-arena>

Images MNIST, CIFAR-10, MicroImageNet, ShapeNetv2

Count 220 hyperparameter-variant neural field datasets

License CC-BY 4.0

Notes Best for studying how SIREN architecture affects FPCA quality. Varies hidden width and training iterations systematically.

COIN-Collection (MS-COCO, HuggingFace)

Paper *Predicting the Encoding Error of SIRENs*, 2024.

Download <https://huggingface.co/datasets/predict-SIREN-PSNR/COIN-collection> (16 GB)

Images MS-COCO, ~287 000 SIRENs

Format HuggingFace dataset; each row contains a base64-encoded state dict.

License MIT

Notes Variable architecture per record. Filter to a single config before use.

Datasets: Shared-backbone + modulation (not standalone)

These datasets store only a small per-image *modulation vector*; the shared backbone must be loaded separately. They are **not** directly usable as standalone INRs but can be adapted.

Functa — CelebA-HQ-64 (DeepMind, ICML 2022)

Paper *From data to functa*, Dupont et al., ICML 2022.

Download gs://dm-functa/celeba_modulations_{64,128,256,512,1024}_latents.npz

GitHub <https://github.com/google-deepmind/functa>

Images CelebA-HQ 64×64, ~27 000 images

Architecture Shared multi-layer SIREN + FiLM modulations; modulation dims: 64/128/256/512/1024

License DeepMind research (check repo)

Notes Only standalone source for CelebA faces as INRs. Per-image data is a modulation vector, not a full MLP. Run `modulation_dataset_writer.py` from the Functa repo to generate fully-independent SIRENs.

MedNF / MedFuncta (550 k+ medical images, Zenodo 2025)

Paper *MedFuncta: A Unified Framework for Learning Efficient Medical Neural Fields*, Friedrich et al., 2025.

Download <https://zenodo.org/records/14898708> (5.2 GB total)

Images 7 MedMNIST subsets (64×64 medical images)

License CC-BY 4.0 (6 datasets); CC-BY-NC 4.0 (DermaNF)

Notes Medical imaging domain. Same modulation paradigm as Functa.

Summary Table

Dataset	Images	# INRs	Arch.	Size	Standalone
DWSNets MNIST	MNIST	70 k	2→32→32→1	~200 MB	Yes
DWSNets FashionMNIST	FashionMNIST	70 k	2→32→32→1	~200 MB	Yes
NFN CIFAR-10	CIFAR-10	60 k	2→64→64→3	~500 MB	Yes
Implicit-Zoo CIFAR-10	CIFAR-10	60 k	2→64→64→3	1.4 GB	Yes
Implicit-Zoo CityScapes	CityScapes	23 k	2→256→⋯→1	18 GB	Yes
Implicit-Zoo ImageNet-1K	ImageNet	1.43 M	2→256→⋯→1	750 GB	Yes
Neural Field Arena	MNIST/CIFAR-10	220 configs	SIREN (variable)	127 GB	Yes
COIN-Collection	MS-COCO	287 k	SIREN (variable)	16 GB	Yes
Functa CelebA-HQ	CelebA-HQ 64px	27 k	Shared+modulation	few MB	No
MedNF	Medical images	550 k+	Shared+modulation	5.2 GB	No

Recommendations for Function-Space PCA

1. **Start here (easiest):** DWSNets MNIST — smallest download, well-studied architecture, 70 k networks, direct `.pt` stacked format. Run: `python fpca.py +experiment=dwsnets_mnist_eulerian`
2. **Natural RGB images:** Implicit-Zoo CIFAR-10 — 60 k standalone SIRENs, 1.4 GB Kaggle download, 3-channel RGB. Run: `python fpca.py +experiment=implicit_zoo_cifar_eulerian`

3. **Architecture sensitivity:** Neural Field Arena (Zenodo) — 220 SIREN configurations per dataset; useful for studying how hidden width / training depth affects FPCA quality.
4. **CelebA faces:** No pre-built standalone INR dataset exists. Options: (a) train your own with Implicit-Zoo generation scripts on CelebA, or (b) use Functa’s modulation vectors and adapt FPCA to modulation space.
5. **Scale:** Implicit-Zoo ImageNet-1K (1.43 M networks) is the largest available, but requires 750 GB storage.

Implementation in this Codebase

Two new classes in `infidictionary/datasets/inr.py`:

DWSNetsINRDataset Loads the single stacked `.pt` format. Batched SIREN evaluation via vectorised `torch.bmm`. Coordinates rescaled $[0, 1]^2 \rightarrow [-1, 1]^2$ internally.

ImplicitZooCIFARDataset Loads individual state-dict `.pt` files from a directory tree. Auto-detects three common key-naming schemes.

Both classes implement the `IrregularDataset` API (same as `CelebADataset`) and are registered in `infidictionary/datasets/__init__.py`.

YAML configs added:

- `conf/function_generator/dwsnets_mnist.yaml`
- `conf/function_generator/implicit_zoo_cifar.yaml`
- `conf/fpca_experiment/dwsnets_mnist_eulerian.yaml`
- `conf/fpca_experiment/implicit_zoo_cifar_eulerian.yaml`